

SHIVA SHANKAR S. Ph.D.

Protein Engineer · Molecular Biologist · Computational Protein Designer
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Protein biochemist with 15 years of experience at the intersection of **computational protein design**, **mass spectrometry-based proteomics**, and **structural biology of proteins**. Expertise in inverse folding (ProteinMPNN, RFdiffusion, Bindcraft), structure-guided engineering, and advanced proteomics workflows. Track record of converting insoluble or poorly characterized proteins into functional variants with biomedical relevance, including a co-invented patent for a glycation-resistant insulin polypeptide.

PROTEIN DESIGN	MASS SPECTROMETRY	STRUCTURAL BIOLOGY
ProteinMPNN · RFdiffusion · Bindcraft · AlphaFold2 · RoseTTAFold · Rosetta	XL-MS · DIA/SWATH · PRM · Top-Down · MaxQuant · Skyline · FragPipe	Crystallography · CD · SEC-MALS · ITC · DLS · AFM · PyMOL · ChimeraX

EDUCATION

Ph.D. in Biological Sciences <i>CSIR-National Chemical Laboratory (CSIR-NCL), Pune, India</i> Thesis: Structural and functional characterization of Meteorin and Meteorin-like proteins Advisor: Dr. Mahesh Kulkarni	Jul 2017 - Dec 2025
M.Sc. in Biotechnology <i>Bharathiar University, India</i>	2009
B.Sc. in Biotechnology <i>Bharathiar University, India</i>	2007

RESEARCH EXPERIENCE

Senior Research Fellow / Project Associate <i>CSIR-National Chemical Laboratory, Pune, India</i>	Jul 2017 - Dec 2025
Project 1: Meteorin and Meteorin-like Protein Characterization - First structural characterization of Meteorin family proteins using AlphaFold2/ColabFold and RoseTTAFold; identified functional receptor-binding regions and high-risk missense variants (JBSD, 2023). - Applied ProteinMPNN-based inverse folding to redesign insoluble mouse Metrnl into a soluble, functionally active variant-manuscript in preparation (first & co-corresponding author). - Developed targeted proteomics assay (PRM) for Metrnl quantification; mapped protein-protein interactions using cross-linking mass spectrometry (XL-MS) integrated with computational modeling.	
Project 2: Glycation-Resistant Insulin Engineering - Engineered a novel glycation-resistant insulin proteoform via de novo solubility-tag design; co-invented Patent WO 2024/194897 A1 (issued Sep 2024). - Designed original in vitro glycation workflow; characterized modification profiles by bottom-up and top-down mass spectrometry. - Optimized recombinant expression and inclusion body refolding systems, achieving significant yield improvements.	
Project 3: SARS-CoV-2 Main Protease Inhibitor Discovery Contributed computational structural analysis and molecular docking to identify multi-target-directed ligands against SARS-CoV-2 Mpro (JBSD, 2021).	
Project Assistant II <i>CSIR-National Chemical Laboratory, Pune, India</i>	Apr 2013 - Jul 2017
Project: Santalene Synthase Structural Characterization (Santalum album) Cloning, mutagenesis, multi-step purification, robotic crystallization, and X-ray diffraction of Santalene synthase; contributed to functional characterization (Sci. Rep., 2015).	
Junior Research Fellow <i>IISER-Pune, India</i>	Jul 2010 - Apr 2013
Project: HIV-1 Fusion Inhibitors & Antimicrobial Lipopeptides Synthesized and characterized γ- and hybrid γ-peptide HIV-1 fusion inhibitors; investigated self-assembling antimicrobial lipopeptides (first-author, J. Med. Chem., 2013).	
Project Trainee <i>National Institute of Immunology (NII), New Delhi, India</i>	Jul 2009 - Apr 2010
Cloned, expressed, and purified HisA, HisB, and HisC2 proteins from <i>M. tuberculosis</i> histidine biosynthesis pathway for crystallographic studies.	

TECHNICAL EXPERTISE

Structural Biology & Crystallography

Protein crystallization (hanging drop, sitting drop, vapor diffusion) · Robotic crystallization systems (Hamilton, Mosquito) · Crystal optimization · X-ray diffraction data collection · Structure prediction (AlphaFold2/ColabFold, RoseTTAFold) · Inverse folding & sequence design (ProteinMPNN, RFdiffusion, Bindcraft) · Homology modeling (Modeller, Schrödinger Prime) · Molecular dynamics simulations (GROMACS, Desmond) · Protein-ligand docking · PyMOL · ChimeraX

Proteomics & Mass Spectrometry

Bottom-up and top-down proteomics · Untargeted and targeted workflows (DDA, DIA/SWATH, PRM) · Targeted assay development · Chemical cross-linking mass spectrometry (XL-MS) for protein interaction mapping · PTM enrichment and analysis (phosphorylation, glycation) · Immunoprecipitation-MS · MALDI · Data analysis: MaxQuant, Skyline, FragPipe, R, Python

Protein Expression, Purification & Characterization

Recombinant expression (E. coli, insect cells, mammalian cells) · Molecular cloning and site-directed mutagenesis · Chromatography (affinity, ion-exchange, gel filtration) · HPLC · FPLC (ÄKTA) · Inclusion body solubilization and refolding · SEC-MALS · Dynamic light scattering (DLS) · Circular dichroism spectroscopy · Isothermal titration calorimetry (ITC) · Atomic force microscopy (AFM)

Molecular & Cell Biology

Mammalian cell culture · CRISPR-based knockout generation · Immunofluorescence · Cell-based assays · PCR / qPCR / RT-PCR · Western blotting · ELISA

PUBLICATIONS & PATENTS

Patent

2024 A glycation-resistant insulin polypeptide and a method of preparing the same thereof

Shankar, S.S.*, Kulkarni, M.J., Banarjee, R., Jathar, S., Rajesh, S.

WO 2024/194897 A1 · Issued Sep 26, 2024

Peer-Reviewed Articles (* First author / Co-corresponding author)

2025 A simple freeze-thaw based method for efficient purification of recombinant human proinsulin from inclusion bodies

Rajesh, S., Jathar, S., Banarjee, R., Sharma, M., Palkar, S., Shankar, S.S.*, & Kulkarni, M.J.

Protein Expression and Purification, 227: 106645

doi.org/10.1016/j.pep.2024.106645

2023 De novo Structure Prediction of Meteorin and Meteorin-Like Protein for Identification of Domains, Functional Receptor Binding Regions, and High-Risk Missense Variants

Shankar, S.S.*, Banarjee, R., Jathar, S.M., Rajesh, S., Ramasamy, S., & Kulkarni, M.J.

Journal of Biomolecular Structure and Dynamics, 42(9): 4522–4536

doi.org/10.1080/07391102.2023.2220804

2023 Mechanistic insights on anserine hydrolyzing activities of human carnosinases

Pandya, V.K., Shankar, S.S., Sonwane, B.P., Rajesh, S., Rathore, R., et al.

Biochimica et Biophysica Acta (BBA)-General Subjects, 1867(3): 130290

doi.org/10.1016/j.bbagen.2022.130290

2021 Discovery of potential multi-target-directed ligands by targeting host-specific SARS-CoV-2 structurally conserved main protease

Joshi, R.S., Jagdale, S.S., Bansode, S.B., Shankar, S.S., et al.

Journal of Biomolecular Structure and Dynamics, 39(9): 3099–3114

doi.org/10.1080/07391102.2020.1760137

2015 Functional Characterization of Novel Sesquiterpene Synthases from Indian Sandalwood, *Santalum album*

Srivastava, P.L., Daramwar, P.P., Krithika, R., Pandreka, A., Shankar, S.S., & Thulasiram, H.V.

Scientific Reports, 5: 10095

doi.org/10.1038/srep10095

2013 Self-Assembly to Function: Design, Synthesis, and Broad-Spectrum Antimicrobial Properties of Short Hybrid *E-Vinylogous* Lipopeptides

Shankar, S.S.*, Benke, S.N., Nagendra, N., Srivastava, P.L., Thulasiram, H.V., & Gopi, H.N.

Journal of Medicinal Chemistry, 56(21): 8468–8474

doi.org/10.1021/jm400884w

Manuscripts in Preparation

In prep. Inverse Folding Design Enables Soluble Expression and Functional Activity of Mouse Meteorin-Like Protein

Shankar, S.S.*, Jathar, S., Sharma, M., Dole, A., Rajesh, S., Banarjee, R., Kulkarni, M.J.

In prep. Characterization of Glycation-Modified Insulin Glargine by Top-Down Mass Spectrometry

Jathar, S., Dole, A., Rajesh, S., Shankar, S.S.*, Kulkarni, M.J.

In prep. Structure-Guided Discovery of RAGE V-Domain Antagonists Targeting AGE- and Ligand-Mediated Inflammaging

Shankar, S.S.*, Jathar, S.M., & Kulkarni, M.J.

AWARDS & RECOGNITION

3rd Prize, Basic Science Category - Chellaram Foundation Diabetes Research Award <i>7th International Diabetes Summit, Pune</i>	2023
Travel Award <i>11th Annual Meeting of Proteomics Society of India</i>	2019
CSIR Senior Research Fellowship <i>Competitive national fellowship for doctoral research at CSIR-NCL</i>	2017
GATE Qualified - Biotechnology (All India Rank 862)	2016
Summer Research Fellowship <i>Indian Academy of Sciences</i>	2008

TEACHING & MENTORSHIP

Instructor - Targeted Proteomics Workshop <i>CSIR Integrated Skill Initiative</i> Designed and delivered curriculum on PRM assay development and data analysis using Skyline, trained researchers from national institutes.	2022
Instructor - Basic to Advanced Proteomics Course <i>CSIR Integrated Skill Initiative</i> Led hands-on sessions from sample preparation to data interpretation using MaxQuant and FragPipe.	2021
Student Mentorship <i>CSIR-National Chemical Laboratory</i> Supervised 8 project students and trained 4 Ph.D. students in molecular biology, protein purification, protein design, and proteomics workflows.	2017-2025
Scientific Peer Review: Analytical Biochemistry; Protein Expression and Purification	

SELECTED CONFERENCES & TRAINING

73rd ASMS Conference on Mass Spectrometry, Baltimore, USA (Jun 2025) · 21st Annual US HUPO Conference, Philadelphia, USA (Feb 2025) · Advanced Proteomics Technologies Workshop, IIT Bombay (Feb 2024) · Monsoon Advanced Proteomics School (MAPS), IIT Bombay (Jun 2023) · 7th International Diabetes Summit, Pune (Mar 2023) · 24th IUCr Congress of Crystallography, India (Aug 2017)

REFERENCES

Dr. Mahesh J. Kulkarni

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Dr. Sureshkumar Ramasamy

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